

Loss	UCSF (Tumor)				LUMIERE (Tumor)				LMSLS (MS)			
	DSC $\uparrow$	HD $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	DSC $\uparrow$	HD $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	DSC $\uparrow$	HD $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$
w/o $\mathcal{L}_{\text{PDE}}$	0.745	26.216	31.359	0.920	0.524	34.211	37.137	0.953	0.690	22.337	29.312	0.898
w/ $\mathcal{L}_{\text{PDE}}$	0.752	25.952	31.364	0.920	0.534	33.672	37.216	0.957	0.696	20.867	29.314	0.900

Table 3: Ablation of PDE-regularized loss across three longitudinal datasets. We compare models trained with and without $\mathcal{L}_{\text{PDE}}$. The model with $\mathcal{L}_{\text{PDE}}$ consistently improves lesion-level metrics while maintaining image-level performance.